

Connecting Data Sources and Building Models

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I began by running the SQL commands within pgAdmin 4 (Figure 1) to create the dataset. This produced the Authors, Genres, Books, Patrons, and the Borrowings tables. Each table was assigned primary and foreign keys to connect the tables. After confirming the tables were created successfully, I linked the database in PostgreSQL into the Power BI environment. Using the Get Data menu, I selected the PostgreSQL Database option and entered my local server and database information (Figure 2). Figure 3 is a screenshot of the connection windows showing how Power BI recognized each table before importing.

Figure 1

SQL Command in pgAdmin 4

```

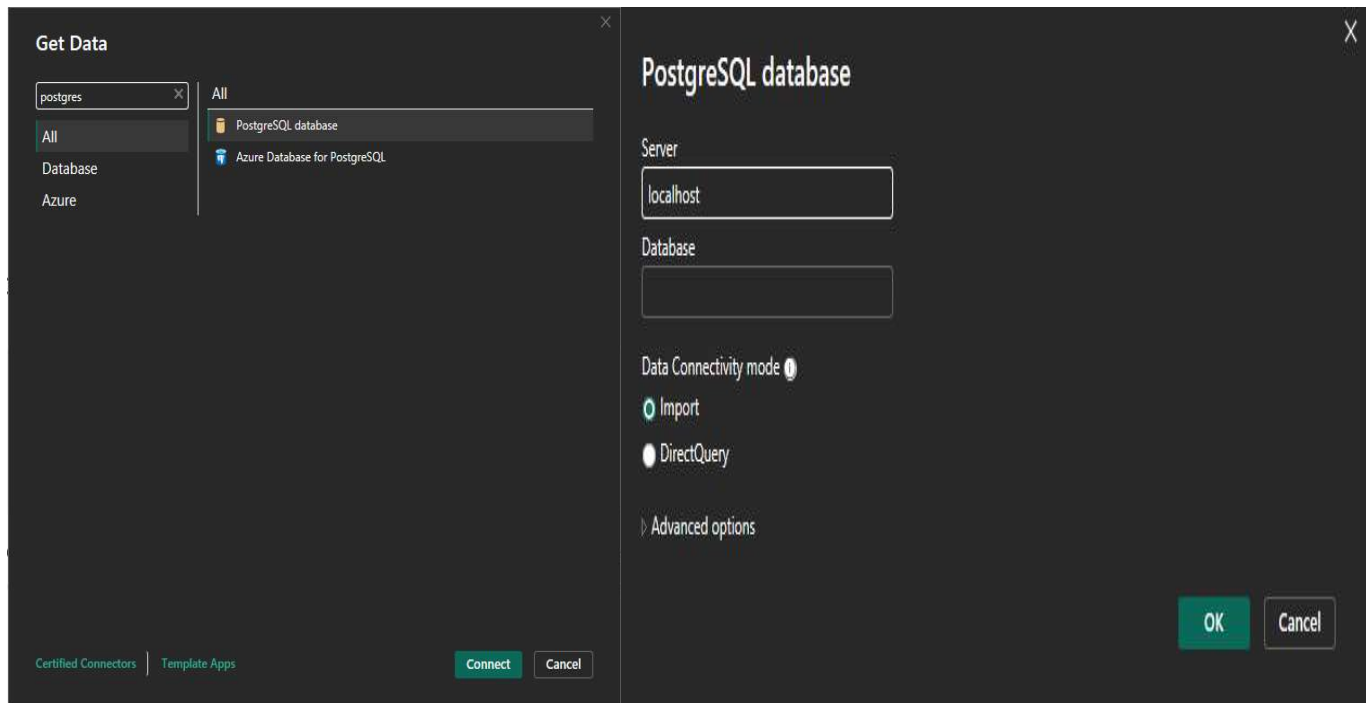
1  CREATE TABLE authors (
2      author_id SERIAL PRIMARY KEY,
3      author_name VARCHAR(100) NOT NULL
4  );
5
6  CREATE TABLE genres (
7      genre_id SERIAL PRIMARY KEY,
8      genre_name VARCHAR(50) NOT NULL
9  );
10
11 CREATE TABLE books (
12     book_id SERIAL PRIMARY KEY,
13     title VARCHAR(200) NOT NULL,
14     author_id INT REFERENCES authors(author_id),
15     genre_id INT REFERENCES genres(genre_id),
16     publication_year INT
17 );
18
19 CREATE TABLE patrons (
20     patron_id SERIAL PRIMARY KEY,
21     patron_name VARCHAR(100) NOT NULL,
22     join_date DATE
23 );
24
25 CREATE TABLE borrowings (
26     borrowing_id SERIAL PRIMARY KEY,
27     patron_id INT REFERENCES patrons(patron_id),
28     book_id INT REFERENCES books(book_id),
29     borrow_date DATE,
30     due_date DATE
31 );
32
33 INSERT INTO authors (author_name) VALUES
34     ('Author A'),
35     ('Author B'),
36     ('Author C');
37
38 INSERT INTO genres (genre_name) VALUES
39     ('Fiction'),
40     ('Non-Fiction'),
41     ('Mystery');
42
43 INSERT INTO books (title, author_id, genre_id, publication_year) VALUES
44     ('Book 1', 1, 1, 2000),
45     ('Book 2', 2, 2, 2015),
46     ('Book 3', 3, 3, 2022);
47
48 INSERT INTO patrons (patron_name, join_date) VALUES
49     ('Patron X', '2020-01-15'),
50     ('Patron Y', '2021-05-20');
51
52 INSERT INTO borrowings (patron_id, book_id, borrow_date, due_date) VALUES
53     (1, 1, '2023-08-15', '2023-09-15'),
54     (1, 2, '2023-08-20', '2023-09-20'),
55     (2, 3, '2023-08-25', '2023-09-25');
56

```

Note. This figure shows the SQL schema used to generate the Authors, Genres, Books, Patrons, and Borrowings table.

Figure 2

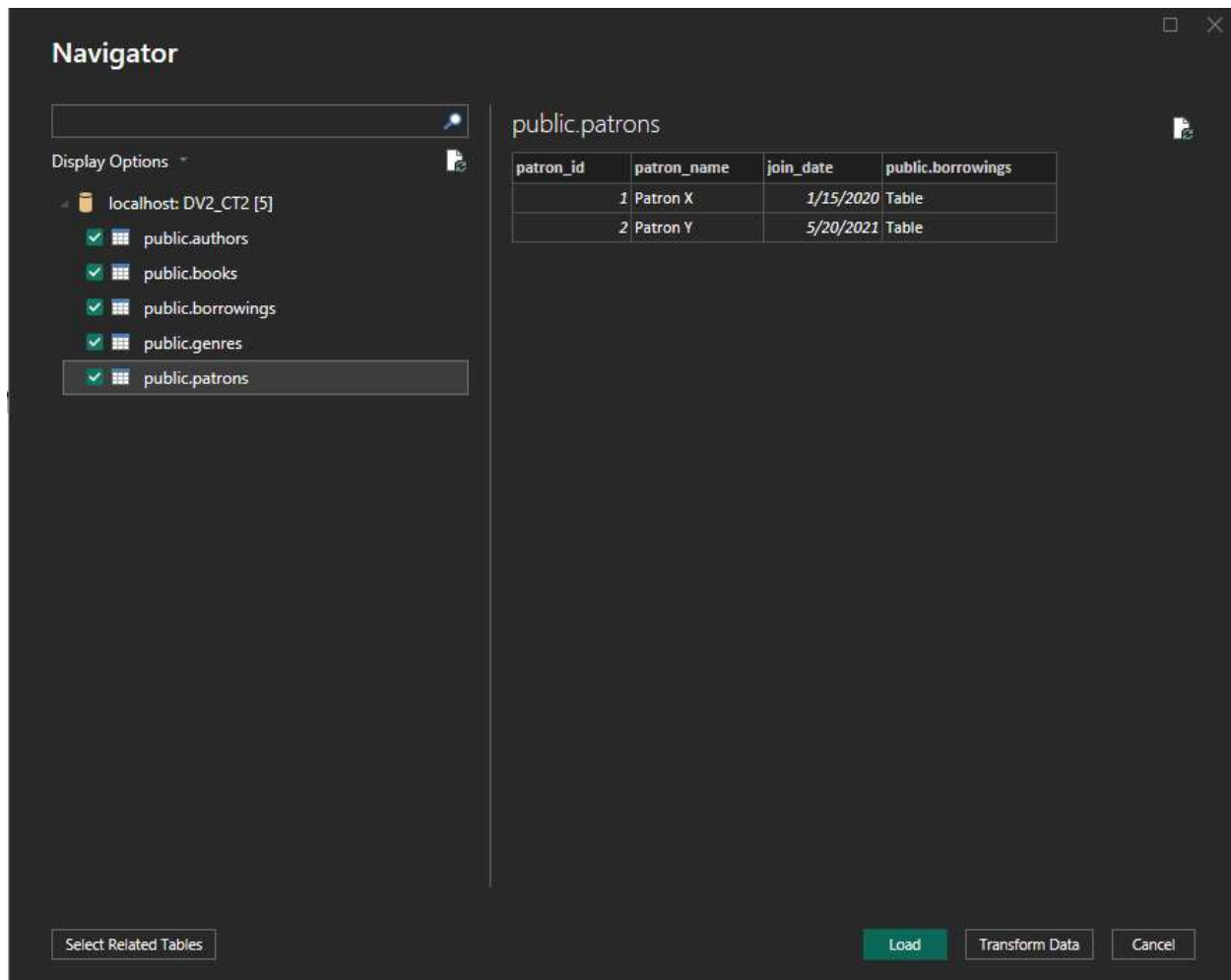
Initial Power BI connection through the Get Data menu



Note. This figure displays the selection of the PostgreSQL Database option used to connect Power BI to the local server. It also indicates which server the database will be pulling from (localhost) and the database itself (DV2_CT2).

Figure 3

Navigator Window displaying Available Table



Note. This figure confirms that Power BI successfully detected all tables before loading them into the model.

Once the tables were loaded, I moved into the Model view to examine the structure of the dataset. I renamed each table to a clear readable title as the word “public” was parsed in the titles when database was transitioned. I then verified if all the relationships were correct.

Unfortunately, Power BI’s model was not correct so I utilized the data modeling features to rectify the cardinalities. Figure 4 demonstrates how I reviewed the primary and foreign key links and confirmed that each relationship reflected the appropriate 1:M structure. I set the cardinality in the following ways:

- authors.author_id → books.author_id (One-to-Many)

- genres.genre_id → books.genre_id (One-to-Many)
- books.book_id → borrowings.book_id (One-to-Many)
- patrons.patron_id → borrowings.patron_id (One-to-Many)

The resulting data model is represented by Figure 5.

Figure 4

Data Model Relationship Editing

Edit relationship ✕

Select tables and columns that are related.

From table
books

author_id	book_id	genre_id	publication_y...	title
1	1	1	2000	Book 1
2	2	2	2015	Book 2
3	3	3	2022	Book 3

To table
authors

author_id	author_name
1	Author A
2	Author B
3	Author C

Cardinality
Many to one (*:1)

Cross-filter direction
Single

☒ Make this relationship active

☐ Apply security filter in both directions

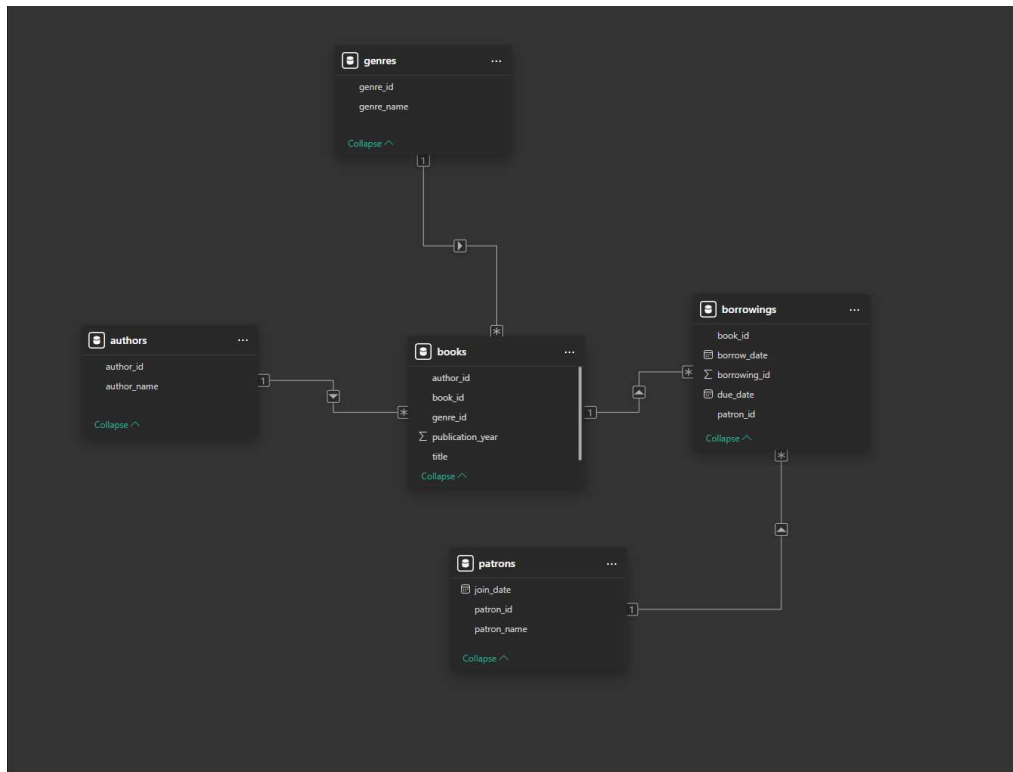
☐ Assume referential integrity

Save **Cancel**

Note. This figure illustrates the verification and correction of the 1:M relationships between the tables.

Figure 5

Complete Data Model Structure in Power BI

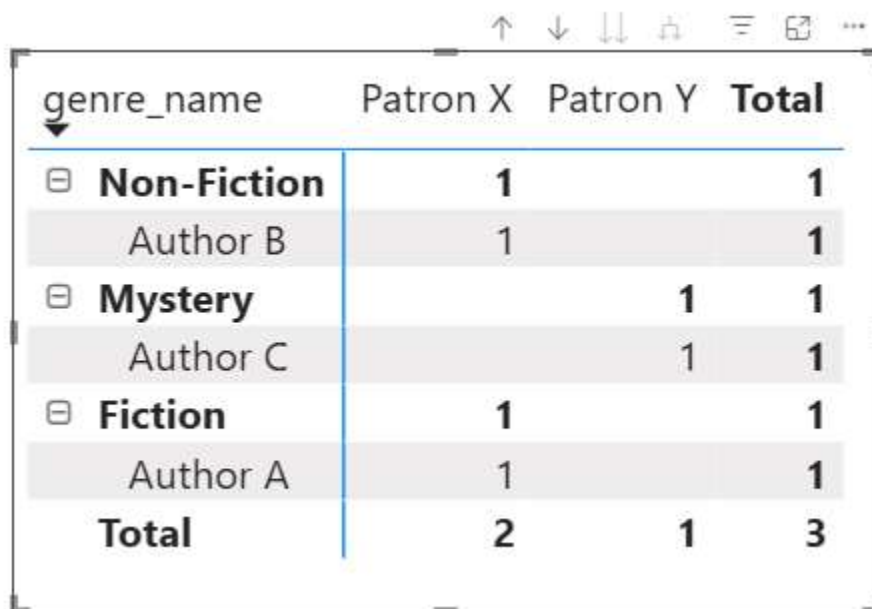


Note. This figure presents the fully configured data model that connects the tables using 1:M across all primary and foreign keys.

To validate that the data model functioned as intended, I created a matrix visualization that used fields utilizing all tables. Figure 6 displays accurate results across genres, authors, patrons, and borrowing counts, which confirms that the relationships were configured correctly and that Power BI can interpret the model without any errors.

Figure 6

Matrix Visualization Validating the Connected Data Model



genre_name	Patron X	Patron Y	Total
⊖ Non-Fiction	1		1
Author B	1		1
⊖ Mystery		1	1
Author C		1	1
⊖ Fiction	1		1
Author A	1		1
Total	2	1	3

Note. This figure demonstrates accurate borrowing counts by genre, author, and patron. This confirms that all relationships are functioning correctly.